

NO	Chemical	Formula	Irritation	Commonly Found In	Potential Health Risks	pH Level	Toxic Level	Composition	Synthesis Method
1	Sodium Lauryl Sulfate (SLS)	C12H25SO4Na	Skin, eyes, respiratory tract	Shampoos, toothpaste, soaps	Skin irritation, allergic reactions, potential organ toxicity	9.0 - 10.5	High	Organic compound	Synthesis from lauryl alcohol and sulfur trioxide
2	Parabens	-	Skin, eyes, respiratory tract	Cosmetics, personal care products, pharmaceuticals	Hormonal disruption, potential carcinogenicity	-	High	Organic compounds	Synthesis as preservatives in various products
3	Phthalates	-	Skin, eyes, respiratory tract	Plastics, personal care products, food packaging	Endocrine disruption, reproductive issues, developmental problems	-	High	Organic compounds	Synthesis as plasticizers and additives in various products
4	Formaldehyde and Formaldehyde-releasing preservatives	CH2O (and others)	Skin, eyes, respiratory tract	Personal care products, household cleaners	Respiratory issues, cancer, skin irritation, headaches	2.2 - 4.0	High	Methanol oxidation or hydrolysis of paraformaldehyde	Oxidation of methanol or hydrolysis of paraformaldehyde
5	Triclosan	C12H7Cl3O2	Skin, eyes, respiratory tract	Antibacterial soaps, toothpaste, cosmetics	Hormonal disruption, antibiotic resistance, environmental damage	8.0 - 9.0	High	Organic compound	Synthesis from chlorinated phenols and chloroacetyl chloride
6	Polyethylene Glycol (PEG) compounds	-	Skin, eyes, respiratory tract	Cosmetics, pharmaceuticals, laxatives	Irritation, allergic reactions, potential contamination with carcinogens	-	High	Organic compounds	Synthesis as industrial chemicals or byproducts
7	Synthetic colors	-	Skin, eyes, respiratory tract	Food, cosmetics, pharmaceuticals	Allergic reactions, hyperactivity in children, potential carcinogenicity	-	High	Organic compounds	Synthesis from petrochemicals
8	BHA (Butylated Hydroxyanisole)	C11H16O2	Skin, eyes, respiratory tract	Food, cosmetics, pharmaceuticals	Hormonal disruption, potential carcinogenicity	-	High	Organic compound	Synthesis from phenol and isobutylene
9	BHT (Butylated Hydroxytoluene)	C15H24O	Skin, eyes, respiratory tract	Food, cosmetics, pharmaceuticals	Hormonal disruption, potential carcinogenicity	-	High	Organic compound	Synthesis from para-cresol and isobutylene
10	Petroleum-based ingredients	-	Skin, eyes, respiratory tract	Cosmetics, personal care products, pharmaceuticals	Skin irritation, allergic reactions, potential contamination with toxins	-	High	Organic compounds	Fractional distillation of crude oil

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11	Toluene	C ₇ H ₈	Skin, eyes, respiratory tract	Paint thinners, adhesives, gasoline	Central nervous system damage, reproductive issues, liver and kidney damage	7.0 - 8.0	High	Aromatic hydrocarbon	Catalytic reforming of crude oil
12	Ethanolamines (MEA, DEA, TEA)	-	Skin, eyes, respiratory tract	Cosmetics, shampoos, cleaning products	Allergic reactions, eye irritation, potential carcinogenicity	-	High	Organic compounds	Synthesis from ammonia and ethylene oxide
13	Retinyl Palmitate (Vitamin A Palmitate)	C ₃₆ H ₆₀ O ₂	Skin, eyes, respiratory tract	Cosmetics, sunscreen	Skin irritation, photosensitivity, potential developmental toxicity	-	High	Organic compound	Synthesis from retinol and palmitic acid
14	Coal Tar	-	Skin, eyes, respiratory tract	Shampoos, soaps, dandruff treatments	Cancer (especially skin), allergic reactions, potential developmental toxicity	-	High	Complex mixture	Distillation of bituminous coal
15	Siloxanes (e.g., Cyclotetrasiloxane, Cyclopentasiloxane)	-	Skin, eyes, respiratory tract	Cosmetics, personal care products, textiles	Hormonal disruption, potential reproductive and developmental toxicity	-	High	Organic compounds	Synthesis as byproduct of silicone production
16	Lead and other Heavy Metals	-	Skin, eyes, respiratory tract, gastrointestinal tract	Contaminated water, food, cosmetics	Neurological damage, developmental issues (especially in children), reproductive problems	-	High	Heavy metals	Natural occurrence or refining of ores
17	Aluminum Compounds	-	Skin, eyes, respiratory tract	Antiperspirants, food additives, cookware	Neurological damage, potential link to Alzheimer's disease	-	High	Inorganic compounds	Mining and refining of aluminum ores
18	Talc	Mg ₃ Si ₄ O ₁₀ (OH) ₂	Skin, eyes, respiratory tract	Baby powder, cosmetics, pharmaceuticals	Respiratory issues, potential carcinogenicity	8.0 - 9.0	High	Mineral	Natural occurrence and mining
19	Hydroquinone	C ₆ H ₆ O ₂	Skin, eyes, respiratory tract	Skin lightening creams, hair dyes, photographic developers	Skin irritation, potential carcinogenicity	-	High	Organic compound	Synthesis from benzene and hydrochloric acid

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20	Oxybenzone	C ₁₄ H ₁₂ O ₃	Skin, eyes, respiratory tract	Sunscreens	Hormonal disruption, potential developmental toxicity	-	High	Organic compound	Synthesis from phenol and benzoyl chloride
21	Formaldehyde Donors	-	Skin, eyes, respiratory tract	Personal care products, household cleaners	Skin irritation, allergic reactions, potential carcinogenicity	-	High	Various compounds	Release of formaldehyde from donor compounds
22	Chlorine	Cl ₂	Skin, eyes, respiratory tract	Disinfectants, water treatment	Respiratory issues, skin irritation, eye damage	4.5 - 5.5	High	Inorganic compound	Electrolysis of sodium chloride solution
23	Ammonia	NH ₃	Skin, eyes, respiratory tract	Cleaning products, fertilizers, refrigerants	Respiratory issues, eye irritation, skin burns	11.0 - 12.0	High	Inorganic compound	Synthesis from nitrogen and hydrogen
24	Phthalates	-	Skin, eyes, respiratory tract, gastrointestinal tract	Plastics, personal care products, food packaging	Endocrine disruption, reproductive issues, developmental problems	-	High	Organic compounds	Synthesis as plasticizers and additives in various products
25	Triclosan	C ₁₂ H ₇ Cl ₃ O ₂	Skin, eyes, respiratory tract, gastrointestinal tract	Antibacterial soaps, toothpaste, cosmetics	Hormonal disruption, antibiotic resistance, environmental damage	8.0 - 9.0	High	Organic compound	Synthesis from chlorinated phenols and chloroacetyl chloride
26	Formaldehyde	CH ₂ O	Skin, eyes, respiratory tract	Household products (e.g., cleaners, cosmetics, adhesives), building materials	Respiratory issues, cancer, skin irritation, headaches	2.2 - 4.0	High	Methanol oxidation or hydrolysis of paraformaldehyde	Oxidation of methanol or hydrolysis of paraformaldehyde
27	Lead	Pb	Skin, eyes, respiratory tract, gastrointestinal tract	Old paint, contaminated soil, some cosmetics, water pipes	Neurological damage, developmental issues (especially in children), reproductive problems	-	High	Heavy metal	Natural occurrence or refining of lead ores
28	Bisphenol A (BPA)	C ₁₅ H ₁₆ O ₂	Skin, eyes, respiratory tract	Plastic products, epoxy resins	Hormonal disruption, reproductive issues, developmental problems	-	High	Organic compound	Synthesis from phenol and acetone

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29	Perfluorooctanoic acid (PFOA)	C ₇ HF ₁₅ O ₂	Skin, eyes, respiratory tract	Non-stick cookware, water-resistant coatings	Cancer (especially liver), developmental issues, immune system suppression	-	High	Organic compound	Synthesis from fluorotelomers
30	Mercury	Hg	Skin, eyes, respiratory tract, gastrointestinal tract	Fish (especially predatory fish), dental fillings, thermometers	Neurological damage, developmental issues (especially in children), kidney damage	-	High	Heavy metal	Mining and refining of cinnabar ore, combustion of coal
31	Arsenic	As	Skin, eyes, respiratory tract, gastrointestinal tract	Contaminated water, some foods, pesticides	Cancer (especially skin, lung, and bladder), cardiovascular issues, gastrointestinal issues	-	High	Metalloid	Natural occurrence or as a byproduct of metal smelting
32	Carbon monoxide	CO	Respiratory tract	Incomplete combustion, vehicle emissions	Headaches, dizziness, death (in high concentrations)	-	High	Inorganic compound	Combustion of carbon-containing fuels
33	Sulfuric acid	H ₂ SO ₄	Skin, eyes, respiratory tract	Industrial processes, batteries, cleaning agents	Burns, respiratory issues, skin corrosion	0.3 - 0.4	High	Inorganic acid	Synthesis from sulfur dioxide and water
34	Hydrochloric acid	HCl	Skin, eyes, respiratory tract	Industrial processes, chemical synthesis	Burns, eye damage, respiratory issues	-	High	Inorganic compound	Synthesis from chlorine and hydrogen
35	Acetone	CH ₃ COCH ₃	Skin, eyes, respiratory tract	Nail polish remover, paint thinner, solvents	Irritation, central nervous system depression, liver and kidney damage	7	High	Ketone	Synthesis from isopropyl alcohol
36	Formalin	CH ₂ O	Skin, eyes, respiratory tract	Disinfectants, embalming fluids	Irritation, respiratory issues, potential carcinogenicity	2.2 - 4.0	High	Solution of formaldehyde in water	Oxidation of methanol or hydrolysis of paraformaldehyde
37	Cadmium	Cd	Skin, eyes, respiratory tract, gastrointestinal tract	Metal plating, electroplating	Kidney damage, lung cancer, bone damage	-	High	Inorganic compound	Synthesis from cadmium and hydrochloric acid

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38	Benzene	C ₆ H ₆	Skin, eyes, respiratory tract, gastrointestinal tract	Tobacco smoke, gasoline, industrial solvents	Cancer (especially leukemia), central nervous system damage, reproductive issues	4.4 - 5.4	High	Aromatic hydrocarbon	Reforming of crude oil or extraction from coal tar
39	Perchloroethylene	C ₂ Cl ₄	Skin, eyes, respiratory tract	Dry cleaning solutions, degreasers	Liver and kidney damage, central nervous system depression, cancer	7.0 - 8.0	High	Organic compound	Synthesis from ethylene and chlorine
40	Carbon tetrachloride	CCl ₄	Skin, eyes, respiratory tract	Cleaning products, solvents	Liver and kidney damage, central nervous system depression, cancer	4.0 - 5.0	High	Organic compound	Synthesis from methane and chlorine
41	Hexavalent chromium	Cr(VI)	Skin, eyes, respiratory tract, gastrointestinal tract	Industrial processes, metal plating	Cancer (especially lung), respiratory issues, gastrointestinal issues	-	High	Inorganic compound	Synthesis from chromium trioxide and sulfur dioxide
42	Methanol	CH ₃ OH	Skin, eyes, respiratory tract	Solvents, antifreeze, fuel	Headaches, dizziness, blindness (in high concentrations)	7.5 - 8.0	High	Alcohol	Synthesis from carbon monoxide and hydrogen
43	Carbon Disulfide	CS ₂	Skin, eyes, respiratory tract	Industrial processes, solvents, pesticides	Neurological damage, reproductive issues, liver and kidney damage	2.0 - 3.0	High	Organic compound	Synthesis from carbon and sulfur
44	Cadmium Chloride	CdCl ₂	Skin, eyes, respiratory tract	Metal plating, electroplating	Kidney damage, lung cancer, bone damage	-	High	Inorganic compound	Synthesis from cadmium and hydrochloric acid